Community Detection on Signed Graphs: A Bayesian MCMC Coupling Approach

Seminar of the MathNet team (Inria Paris)

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Inria Paris, Room Paul Erdös (A115) – 2 p.m. Monday 15 June 2026

Outline

01
Motivation & the Sankararaman–Baccelli Model

Haplotype Assembly

1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

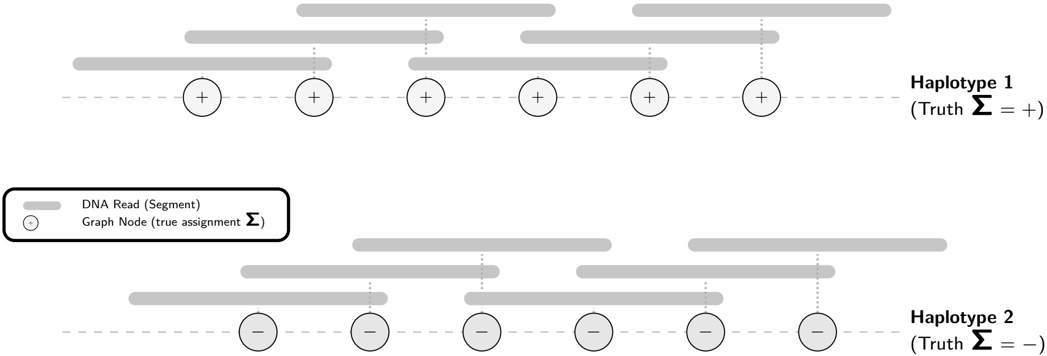
Haplotype 1
(Truth $\boldsymbol{\Sigma} = +$)

Haplotype 2
(Truth $\boldsymbol{\Sigma} = -$)

Haplotype Assembly

2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.

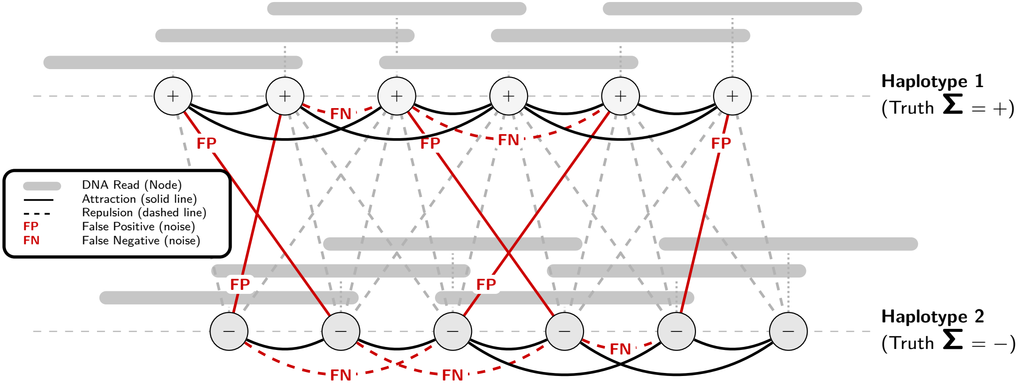


Haplotype Assembly

3. Modeling as a Signed Graph

Nodes = reads   Edges = overlaps

Goal: Reconstruct the true haplotype assignments $\boldsymbol{\Sigma}$ from this observed signed graph.



Motivation: Signed Community Detection

Community Detection on Signed Graphs ($K = 2$)

- We aim to partition nodes into $K = 2$ communities: $\boldsymbol{\Sigma}_i \in \{+1, -1\}$.
- Observed links represent **attractions** (solid lines) and **repulsions** (dashed lines).
- Under the hood, this is a **Stochastic Block Model** (SBM) setup.

The Geometric Challenge

- Nodes are embedded in space (e.g., genomic positions on $\mathbb{Z}$ or $\mathbb{R}^d$).
- Edge probabilities depend directly on the distance $r = \|x_i - x_j\|$:

$$\mathbb{P}(\text{attractive edge} \mid \text{same community}) = f_{\text{in}}(r)$$
$$\mathbb{P}(\text{repulsive edge} \mid \text{different communities}) = f_{\text{out}}(r)$$

- **All the difficulty (and the interest!) comes from this geometric dependency.**

The Sankararaman–Baccelli Model (GSBM)

Model Definition [1]

- **Spatial Nodes** V_n: Hom. Poisson PP $\mathcal{X}$ in $\mathbb{R}^d$ [2] restrict. to a n-volume window.
- **Latent Communities**: Uniform assignments $\Sigma_i \in \{+1, -1\}$ $(K = 2)$.
- **Geometric Overlaps**: Observed connection $e = \{i, j\} \in H$ with probability:

$$\mathbb{P}(e \in H \mid X, \Sigma) = \begin{cases} f_{\mathrm{in}}(r_e) & \text{if } \Sigma_i = \Sigma_j \text{ (attraction)}, \\ f_{\mathrm{out}}(r_e) & \text{if } \Sigma_i \neq \Sigma_j \text{ (repulsion)}, \end{cases}$$

with distance $r_e = \|X_i - X_j\|$ and connecting functions $f_{\mathrm{in}}(r) \geq f_{\mathrm{out}}(r)$.

Goal: Weak Recovery

- Find an algorithm τ (independent of Σ) reconstructing assignments:

$$\lim_{n \to \infty} \mathbb{E}[\mathrm{ov}_n(\Sigma, \tau)] \geq \frac{1}{2} + \eta \quad \text{for some } \eta > 0,$$

using the **Community Overlap**: $\mathrm{ov}_n(\Sigma, \tau) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}_{\{\tau_i = \pi(\Sigma_i)\}}$.

Gibbs Model on Signed Graphs

Attractive and Repulsive Edges

We partition the potential edges $E = F \dot{\cup} \tilde{F}$:

- F: **Attractive** (ferromagnetic) edges, with weight $W_e > 0$
- $\tilde{F}$: **Repulsive** (antiferromagnetic) edges, with weight $W_e < 0$

Signed Gibbs Measure [3]

The probability of a label configuration $\sigma \in \{+1, -1\}^{V_n}$ is $\mu(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$, where the energy $U(\sigma)$ penalizes unsatisfied edges:

$$U(\sigma) = \sum_{e=\{i,j\} \in F} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{e=\{i,j\} \in \tilde{F}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}$$

The magnitudes $|W_e|$ determine the strength of each constraint.

From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- With fixed spatial positions X, all pairwise distances $r_e = \|X_i - X_j\|$ are known.
- Under a uniform prior on Σ, the **posterior** distribution given the observed graph H is:

$$\mathbb{P}(\Sigma = \sigma \mid H, X) \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

- Choosing weights $W_e(H)$ as below:

$$W_e(H) = \begin{cases} \log\left(\dfrac{f_{\mathrm{in}}(r_e)}{f_{\mathrm{out}}(r_e)}\right) > 0 & \text{if } e \in H \text{ (observed overlap } \in F), \\ -\log\left(\dfrac{1 - f_{\mathrm{out}}(r_e)}{1 - f_{\mathrm{in}}(r_e)}\right) < 0 & \text{if } e \notin H \text{ (absent overlap } \in \tilde{F}). \end{cases}$$

- ... yields the exact correspondence:

$$\mathbb{P}(H \mid X, \sigma) \propto e^{-U(\sigma)}$$

From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- With fixed spatial positions X, all pairwise distances $r_e = \|X_i - X_j\|$ are known.
- Under a uniform prior on Σ, the **posterior** distribution given the observed graph H is:

$$\boldsymbol{\mathbb{P}(\Sigma = \sigma \mid H, X) = \mu(\sigma)} \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

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The Percolation Obstruction

Necessary Condition for Weak Recovery [1]

- Intuitively, if the graph is composed of many tiny, isolated components, no global community structure can be reconstructed.
- Information can only flow globally if the graph **percolates** (possesses a giant/infinite component).

Theorem (Sankararaman & Baccelli [1])

Percolation of the graph with edge probability

$$p_e = f_{\mathrm{in}}(r_e) - f_{\mathrm{out}}(r_e)$$

*is a **necessary condition** for weak recovery. If this independent bond percolation does not yield a giant component, weak recovery is mathematically impossible.*

- Our goal: Retrieve and extend this result using a unified **Bayesian MCMC coupling** framework.

02
The Bayesian Framework & Coupling

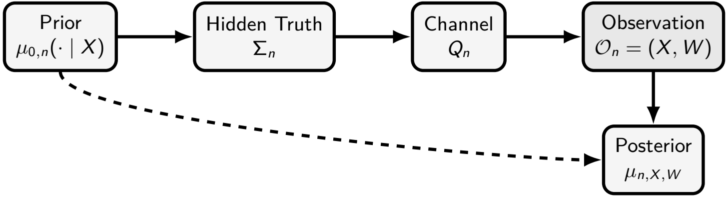
The General Bayesian Model

Definition

General Bayesian Community Detection Model

1. **Features**: $X_n \in \mathcal{X}_n$ with distribution ν_n.
2. **Prior**: $\mu_{0,n}(\mathrm{d}\sigma \mid x)$ over $\mathcal{C}_n = \mathcal{L}_K^{V_n}$.
3. **Channel**: $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$.

The joint law is $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x)\mu_{0,n}(\mathrm{d}\sigma \mid x)Q_n(\mathrm{d}w \mid x, \sigma)$.



Posterior Resampling (Nishimori Identity)

Replacing Truth by a Replica

- Since Σ_n is hidden, we cannot compute the overlap $\mathrm{ov}_n(\Sigma_n, \tau_n)$ directly in an algorithm.
- However, under the true model, the hidden truth and a sample from the posterior are exchangeable.

Theorem (Posterior Resampling / Nishimori Identity [4])

Let τ_n be any community detection algorithm. Let $\Sigma_n^{(1)} \sim \mu_{n,X_n,W_n}$ be a replica drawn independently from the posterior. For any bounded test function Ψ_n:

$$\mathbb{E}\left[\Psi_n(X_n, W_n, \Sigma_n, \tau_n)\right] = \mathbb{E}\left[\Psi_n(X_n, W_n, \Sigma_n^{(1)}, \tau_n)\right].$$

Corollary

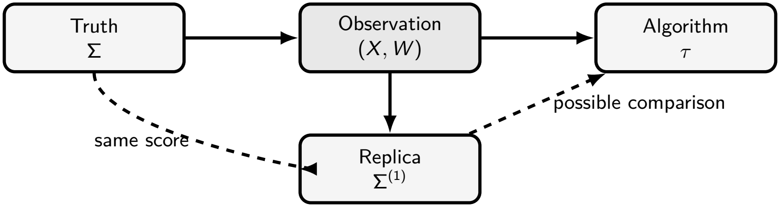
Overlap Identity For any algorithm τ_n and threshold $s \in [0,1]$:

$$\mathbb{P}\left[\mathrm{ov}_n(\Sigma_n, \tau_n) \geq s\right] = \mathbb{P}\left[\mathrm{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq s\right].$$

Invariant Coupling

Markov Chain Dynamics as a Source of Coupling

- To study the overlap, we need a way to construct the posterior replica $\Sigma^{(1)}$ starting from the ground truth Σ.
- Let $M_{n,x,w}$ be a Markov transition kernel that leaves the posterior $\mu_{n,x,w}$ invariant.
- If we initialize the chain at the hidden truth Σ, and perform **exactly one step** of the dynamics to obtain $\Sigma^{(1)} \sim M_{n,X,W}(\Sigma, \cdot)$, then $\Sigma^{(1)}$ is still distributed according to the posterior!



03
MCMC Dynamics & Performance Bounds

MCMC Dynamics for Signed Gibbs Measures

1. Single-Spin Dynamics (Glauber / Metropolis–Hastings)

- Select a node at random and update its label according to its local neighbors.
- Extremely slow in the presence of frustration: local updates cannot cross energy barriers.

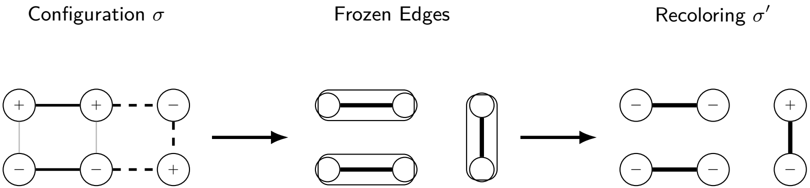
2. Swendsen–Wang (SW) Signed Cluster Dynamics [5, 6]

- Update whole clusters of vertices simultaneously to speed up mixing.
- **Freezing Step**: For each edge $e = \{i, j\}$:
  - If e is satisfied ($W_e > 0$ and $\sigma_i = \sigma_j$, or $W_e < 0$ and $\sigma_i \neq \sigma_j$), freeze it with probability:

$$p_e^{\mathrm{gel}} = 1 - e^{-|W_e|}.$$

  - If e is unsatisfied, do not freeze it ($p_e^{\mathrm{gel}} = 0$).
- **Recoloring Step**: Flip each frozen cluster independently with probability $1/2$.

One Step of Signed Swendsen–Wang



- Edges in bold represent satisfied interactions that are randomly frozen.
- Clusters are then flipped independently.

Coupling & the Sankararaman–Baccelli Bound

Gel Probability under the Generative Model

Let's run one step of SW starting from the hidden truth Σ. What is the probability that an edge e is frozen?

- **Case 1:** $\Sigma_i = \Sigma_j$. Edge is present with probability $f_{\text{in}}(r_e)$, and has weight $W_e = \log(f_{\text{in}}/f_{\text{out}}) > 0$.

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e)\left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e)\left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

- **Case 2:** $\Sigma_i \neq \Sigma_j$. Edge is absent with probability $1 - f_{\text{out}}(r_e)$, and has weight $W_e = -\log((1 - f_{\text{out}})/(1 - f_{\text{in}})) < 0$.

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e))\left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

Corollary

*Independent Bond Percolation Under the true generative model, the graph of frozen edges is **exactly** an independent bond percolation with edge probability:*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

High-Order Triangular Dynamics

The Frustration Obstruction

- Standard edge-by-edge SW freezes satisfied edges independently.
- In frustrated systems, this leads to conflicts and blocks dynamics due to excessive freezing.
- **Idea**: Introduce correlations on the freezing mechanism using higher-order cliques (**triangles**) to freeze as little as possible.

3. Triangular Cluster Dynamics

- Transfer edge energy to triangles: $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$.
- **Attractive Triangle** ($+ + +$ or $- - -$): Freeze all 3 edges (prob $a_u = 1 - e^{-2u}$) or none (prob $b_u = e^{-2u}$).
- **Repulsive Triangle** (2 satisfied edges, 1 unsatisfied): Freeze exactly one of the two satisfied edges with probability $a_u/2$. This keeps the system in a low energy state with minimal freezing.

Triangular Dynamics Freezing Rules

Attractive Triangle (Isotropic)					
Configuration	$\varnothing$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+,+,+)$	b_u	0	0	0	a_u
Other	1	0	0	0	0

Repulsive Triangle (Isotropic)					
Configuration	$\varnothing$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+,+,+)$	1	0	0	0	0
$(+,+,-)$	b_u	0	$a_u/2$	$a_u/2$	0

Table: Freezing rules for triangular dynamics ($a_u = 1 - e^{-2u}$, $b_u = e^{-2u}$). By construction, these rules satisfy local reversibility and leave the posterior distribution invariant.

Strict Improvement on the Triangular Grid

Homogeneous Model ($f_{\mathrm{in}} = p$, $f_{\mathrm{out}} = 1 - p$)

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- **Edge Dynamics (Standard SW)**: Gelled graph is independent edge percolation with parameter $p_{\mathrm{edge}} = 2p - 1$. Critical threshold: $p_c^{\mathrm{edge}} = \frac{1}{2} + \sin(\frac{\pi}{18}) \approx 0.673$.
- **Triangular Dynamics**: Gelled graph yields correlated triangle percolation. Critical threshold: $p_c^{\triangle} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192$.

Corollary

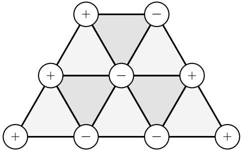
Strictly Better Impossibility Bound [7] Since $p_c^{\mathrm{edge}} < p_c^{\triangle}$, for any

$$p \in [0.673\,;\, 0.7192[\,,$$

standard edge dynamics percole (no conclusion), whereas triangular dynamics do not. Hence, we prove that weak recovery is impossible in this wider range of parameters!

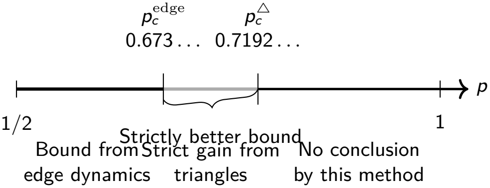
Comparison of Percolation Thresholds

Triangular Grid



White triangles used by triangular dynamics

Weak Recovery Impossibility Bounds



04
Open Questions & Conclusion

Open Questions

1. **Extension to Other Lattices and Higher Dimensions**
   - What are the thresholds on the Kagome, FCC, HCP, or Hyper-cubic lattices?
   - Can we define tetrahedral or higher-order simplex dynamics to obtain even tighter bounds?
2. **Multi-Community case ($K > 2$)**
   - How does the cluster dynamics extend when the spins take K values?
   - What is the role of POTTS-like percolation in the coupling?
3. **Sufficiency of the Percolation Condition**
   - The percolation threshold provides a necessary condition (impossibility).
   - Is percolation also sufficient for the existence of a reconstruction algorithm?

Bibliography I

[1] Abishek Sankararaman and François Baccelli. "Community Detection on Euclidean Random Graphs". In: *Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)* (2018), pp. 2181–2200. DOI: 10.1137/1.9781611975031.142.

[2] François Baccelli and Bartłomiej Błaszczyszyn. *Stochastic Geometry and Wireless Networks, Volume I - Theory*. Vol. 1. Foundations and Trends in Networking. NoW Publishers, 2009, p. 150. DOI: 10.1561/1300000006.

[3] Jörg Reichardt and Stefan Bornholdt. "Statistical mechanics of community detection". In: *Phys. Rev. E* 74 (1 July 2006), p. 016110. DOI: 10.1103/PhysRevE.74.016110.

[4] Hidetoshi Nishimori. "Exact results and critical properties of the Ising model with competing interactions". In: *Journal of Physics C: Solid State Physics* 13.21 (1980), pp. 4071–4076. DOI: 10.1088/0022-3719/13/21/012.

[5] Robert H. Swendsen and Jian-Sheng Wang. "Nonuniversal critical dynamics in Monte Carlo simulations". In: *Phys. Rev. Lett.* 58 (2 Jan. 1987), pp. 86–88. DOI: 10.1103/PhysRevLett.58.86.

Bibliography II

[6] Robert G. Edwards and Alan D. Sokal. "Generalization of the Fortuin-Kasteleyn-Swendsen-Wang representation and Monte Carlo algorithm". In: *Phys. Rev. D* 38 (6 Sept. 1988), pp. 2009–2012. DOI: 10.1103/PhysRevD.38.2009.

[7] Louis Hauseux, Nahuel Soprano-Loto, and Konstantin Avrachenkov. "Higher-order Monte Carlo cluster dynamics for community detection in Euclidean graphs". In: *Asilomar Conference on Signals, Systems, and Computers 2025*. Pacific Grove (CA), United States, Oct. 2025. URL: https://inria.hal.science/hal-05267074.

Merci.